

Dynamic Assembly Line Supervision Using PLC, SCADA, and IoT Technologies

Fatima MOKHLISS
LASTIMI Lab., EST Salé
Mohammed V University in Rabat
Morocco
mokhliss.fatimal@gmail.com

Mohammed Hicham ZAGGAF
LSSDIA Lab., ENSET of Mohammedia
Hassan II University
Morocco
zaggaf@enset-media.ac.ma

El Ayachi CHATER
LASTIMI Lab., EST Salé
Mohammed V University in Rabat
Morocco
chater.el@gmail.com

Abdelmajid BYBI
MEAT Team, EST Salé
Mohammed V University in Rabat
Morocco
abdelmajid_bybi@hotmail.fr

Abstract— In this study, we present the design and implementation of an automated dynamic assembly line to enhance productivity, reduce downtime, and improve the quality of assembly processes by integrating multiple advanced automation technologies. This study aims to demonstrate the feasibility of automating industrial assembly lines, significantly improving operational efficiency and product quality. This setup links the PLC, HMI, SCADA system, and Raspberry Pi to the remote view. The chat among these parts occurs using the OPC UA and MQTT rules employing KepServerEX and Node-RED. In this manner, the plan can ensure easy data sharing, which helps in real-time monitoring and management. A test and check were conducted to confirm how well the system works in collecting information, handling it, and providing remote access to mobile devices.

Keywords— *Dynamic assembly line, PLC, SCADA, HMI, MQTT, Node-RED, OPC UA, Raspberry Pi, Industrial Automation.*

I. INTRODUCTION

Industrial automation has become the basis of the modern manufacturing environment, which ensures that companies gain more productivity, lower costs, and better overall product quality. Concerning that, assembly line operations have been very greatly automated by everything that has come with advanced control systems, sensors, and data acquisition technologies. Turned old manufacturing methods into new ones that would be much more efficient, reliable, and flexible towards changing production demands. This paper presents the design and implementation of an automated dynamic assembly line with a proper contribution from modern technology tools such as programmable logic controllers (PLCs), human-machine interfaces (HMI), supervisory control and data acquisition (SCADA) systems, and the Internet of Things (IoT) [1].

Typically, manual workers on a production line can cause imbalances, high production costs, and even low-quality end products. Therefore, automation is required to balance the desire to optimize the production process, reduce human errors, and add up to greater throughput. In addition, automating repetitive and labor-intensive activities would enable manufacturers to achieve sharper precision cycles that are faster and more efficient overall. This study addresses these issues by proposing

an all-inclusive automated solution that combines several technologies aimed at forming a T-efficient and seamless production system.

The main goal of this study was to plan and implement an automated changing assembly line using PLCs, HMIs, and SCADA systems. In addition, the project aims to merge IoT technology, such as Raspberry Pi, for up-to-the-minute data gathering and faraway watching. Therefore, this mix allows for quick checking, predictive upkeep, and smart choices based on information, making the system even better. The document also aims to highlight the safety and growth key points, ensuring that the setup can change for future production wants and work steadily in different situations [2,3].

This concerns the automation of a dynamic assembly line for cable assembly, designing electrical cabinets, programming PLCs, developing HMIs, and integrating an SCADA system. Further, connected work supports IoT for remote monitoring and control, demonstrating how these technologies can collaboratively function to create a robust and intelligent Automation Solution. The design and implementation results provide an internal view of how industrial automation can leverage manufacturing processes to transform them into modern production environments [4].

The remainder of this paper is organized as follows. Section II presents the architecture and communication methodology of the proposed system. Section III provides details on the configuration of the PLCs with the SCADA system and the MQTT broker. Section IV discusses the simulation results and outlines the possible challenges related to cybersecurity threats. Finally, Section V concludes the study and offers perspectives for future work.

II. SYSTEM ARCHITECTURE AND METHODOLOGY

A. System Overview

The automated assembly line is designed in modular form, wherein each workstation performs a particular function. The organization includes a conveyor belt, sensors, actuators, and control system. Modularity provides flexibility and scalability such that the system can be easily expanded or modified to meet

other (future) production requirements. The conveyor belt carries parts between the workstations, whereas the sensors and actuators monitor and control the assembly process at strategically chosen locations. PLCs, HMIs, SCADA supervision, and IoT control systems should be integrated into a seamless framework for real-time operation monitoring. Thus, each component plays a distinct role in ensuring operational efficiency and process optimization.

- *Siemens S7-1200 PLC*: Controls conveyor movement, manages workstation tasks, and processes sensor inputs.
- *Mitsubishi FX3GE PLC*: Acquires real-time motor condition data, specifically, temperature and vibration measurements.
- *Sensors*: Proximity sensors, thermocouples, and vibration sensors are used to monitor various aspects of the assembly process such as component presence, temperature, and machine vibration.
- *Actuators*: Motors, relays, and contactors are used to control the assembly line, ensuring precise execution of the assembly process.
- *SCADA System*: This allows the supervision of all PLCs (Siemens and Mitsubishi), enabling centralized monitoring and alarm management.
- *HMI*: Provides graphical operator control, allowing process parameter adjustments and system status visualization.
- *Raspberry Pi*: As an IoT gateway, it allows the publishing of real-time sensor data to mobile dashboards via MQTT protocol.

For clarity, Fig.1. illustrates the overall system architecture, including the communication pathways between components.



Fig. 1. System architecture

B. Communication Methods

A critical aspect of industrial automation is guaranteeing efficient communication between controllers, supervisory

systems, and remote-monitoring interfaces. This section details the networking protocol approach, which ensures seamless data flow between all the system components.

PLC-HMI communication relies on PROFINET, a high-speed industrial Ethernet protocol, enabling the real-time exchange of commands and process variables between the Siemens S7-1200 PLC and GT Designer3 HMI.

The SCADA system (WinCC Explorer) was configured to simultaneously supervise both Siemens and Mitsubishi PLCs using an open-platform communication unified architecture (OPC UA). KepServerEX V6 acts as a data gateway, retrieves data from both PLCs, and sends them to WinCC Explorer for centralized monitoring [5].

To facilitate remote monitoring, an MQTT-based communication setup was implemented. The Raspberry Pi functions as a message queuing telemetry transport (MQTT) broker, forwarding real-time process data from the PLCs to a web-based dashboard, which allows operators to access key system parameters via mobile devices. The resulting multi-protocol approach ensures data integrity, security, and accessibility [6].

Fig.2. depicts the network communication architecture, illustrating the data flow between all PLCs, SCADA, and IoT components.

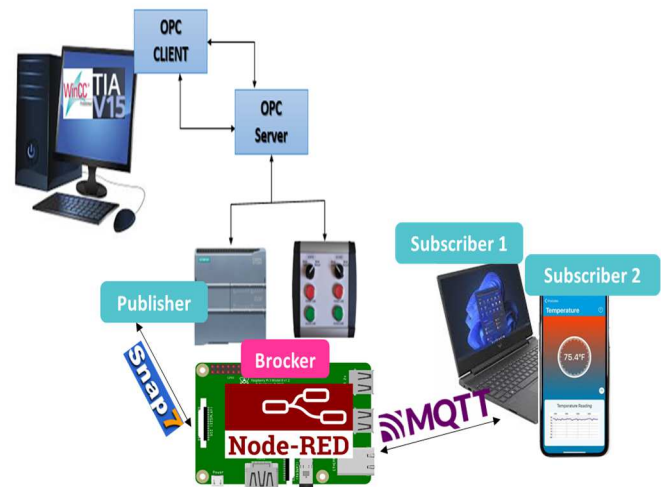


Fig. 2. Network communication architecture

C. Software Tools Description

The system integrates several industrial and IoT-based software applications to enable seamless control, supervision, and remote accessibility [7].

- *TIA Portal*: This is used to program the Siemens S7-1200 PLC using the Ladder Logic language (LAD), configures PROFINET settings, and establishes I/O mapping.
- *GX Works 2*: utilized for Mitsubishi FX3GE PLC programming to handle temperature and vibration sensor data acquisition.

- *GT Designer3*: is used to create an intuitive operator interface for real-time process visualization and manual control.
- *WinCC Explorer*: This was used to monitor all PLCs (Siemens and Mitsubishi) within a single platform, which allowed integration with the KepServerEX V6 OPC UA for real-time process data acquisition.
- *Node-RED*: configured on a Raspberry Pi to process MQTT messages and display real-time sensor readings on any mobile-accessible web dashboard.

For clarity, Fig.3. shows the designed SCADA interface, which allows for the visualization of real-time process data from either Siemens or Mitsubishi PLCs.

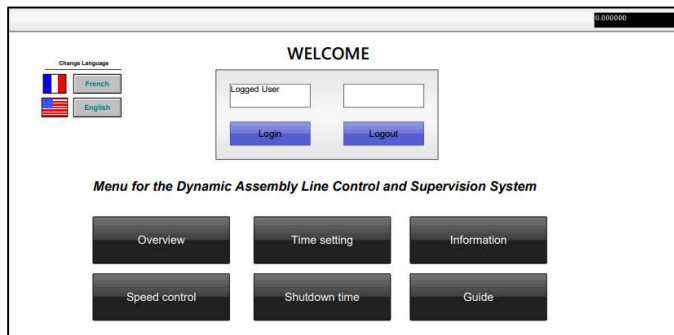


Fig. 3. SCADA interface in WinCC Explorer.

III. IMPLEMENTATION AND TESTING

The implementation of a multi-PLC automation system involves hardware configuration, software development, SCADA integration, IoT-based remote monitoring, and validation testing. This section provides details on how the Siemens S7-1200 PLCs, Mitsubishi FX3GE PLCs, SCADA system (WinCC Explorer), MQTT broker (Raspberry Pi), and Node-RED could be configured to achieve the designed seamless automation and supervision [8].

A. PLC Programming and Control System Development

The PLC programming phase focuses on designing the control logic for assembly line automation and motor condition monitoring. Each PLC is programmed with dedicated logic to handle the respective tasks.

- *Siemens S7-1200 PLC (Assembly Line Automation)*: This system is programmed using Ladder Logic (LAD) in the TIA Portal to control the movement of conveyors, automated workstations, and process inputs from sensors. Interfacing this PLC with proximity sensors and actuators as well as emergency stop buttons was performed through digital and analog I/O for correct material handling and safe operation. The integration of speed control using the Altivar 320 Variable Frequency Drive (VFD) comes next because it is required to provide dynamic conveyor speed variations. The PLC provides analog control signals to the VFD over the 0-10V

range for precise speed regulation based on feedback from the mentioned sensors, as well as reference operator inputs through the HMI [9,10]. Interestingly, safety interlocks are included to manage emergency stops and fault conditions, making it possible to ensure operational security. The real-time visualization of the PLC program structure in the TIA portal is illustrated in Fig. 4.

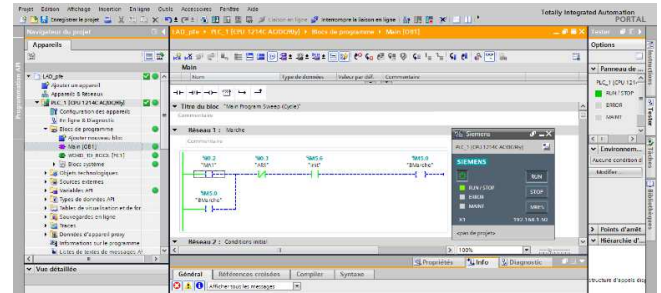


Fig. 4. PLC program structure in the TIA Portal.

- *Mitsubishi FX3GE PLC (Motor Condition Monitoring)*: Programmed using GX Works 2, with a focus on motor condition monitoring through temperature and vibration sensor integration. A J-type thermocouple and vibration sensor were connected to the PLC, allowing a continuous assessment of the motor performance. PLC utilizes analog-to-digital conversion (ADC) to process sensor signals by applying threshold-based alerts to abnormal readings [11]. The collected data can then be transmitted to the WinCC Explorer (SCADA) system via OPC UA communication, enabling both centralized monitoring and predictive maintenance. A diagram illustrating the sensor connections to the Mitsubishi PLC is shown in Fig. 5.

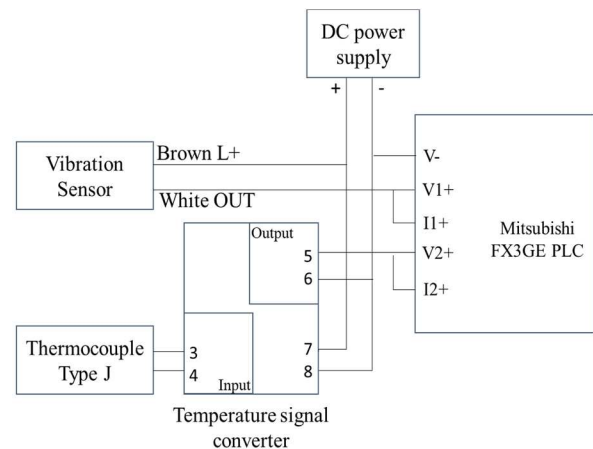


Fig. 5. Sensor connections to the Mitsubishi PLC.

The combination of both PLCs ensures efficient control of the assembly line while continuously monitoring the motor health state, which prevents unexpected failures and reduces maintenance costs.

B. SCADA and Multi-PLC Integration

To enable real-time visualization, data logging, and centralized control, an SCADA system (WinCC Explorer) was configured to supervise both Siemens and Mitsubishi PLCs. This task was performed using the following steps.

- **OPC UA Configuration via KepServerEX V6:** This is a crucial step in establishing a data exchange between PLCs and SCADA. OPC UA tags were created in KepServerEX, representing the sensor values, conveyor speed, and workstation status, along with all health parameters for the motor. These tags were exported from the TIA Portal (for Siemens PLCs) and GX Works 2 (for Mitsubishi PLCs) to KepServerEX so that real-time data could be fetched. The SCADA system can then be configured to subscribe to these OPC UA tags for easy integration. A screenshot of the OPC UA tag configuration in KepServerEX is shown in Fig. 6 for clarity.

Item ID	Data Type	Value	Timestamp	Quality	Update Count
SIEMENS PLC S7-1200 Value_HMI3	Long	0	20:10:20.748	Good	1
SIEMENS PLC S7-1200 Value_HMI2	Long	0	20:10:20.748	Good	1
SIEMENS PLC S7-1200 Value_HMI1	Long	0	20:10:20.748	Good	1
SIEMENS PLC S7-1200 Value_de_corr	DWord	0	20:10:20.748	Good	1
SIEMENS PLC S7-1200 Value_de_corr	DWord	0	20:10:20.748	Good	1
SIEMENS PLC S7-1200 Value_de_corr	DWord	0	20:10:20.748	Good	1
SIEMENS PLC S7-1200 Value_de_corr	DWord	0	20:10:20.748	Good	1
SIEMENS PLC S7-1200 V_vet	Boolean	0	20:10:20.748	Good	1
SIEMENS PLC S7-1200 V_vet	Boolean	1	20:10:20.748	Good	1
SIEMENS PLC S7-1200 V_vet	Boolean	0	20:10:20.748	Good	1
SIEMENS PLC S7-1200 temp_alarm	String	00_0H_0H_0H_0H	20:10:20.748	Good	1
SIEMENS PLC S7-1200 temp_alarm	String	00_0H_0H_0H_0H	20:10:20.748	Good	1
SIEMENS PLC S7-1200 temp_alarm	String	00_0H_0H_0H_0H	20:10:20.748	Good	1
SIEMENS PLC S7-1200 Tag_42	Boolean	0	20:10:20.752	Good	1
SIEMENS PLC S7-1200 Tag_41	Boolean	0	20:10:20.752	Good	1
SIEMENS PLC S7-1200 Tag_40	Short	0	20:10:20.748	Good	1
SIEMENS PLC S7-1200 Tag_39	Float	0	20:10:20.748	Good	1
SIEMENS PLC S7-1200 Tag_38	Short	0	20:10:20.748	Good	1
SIEMENS PLC S7-1200 Tag_37	Short	0	20:10:20.748	Good	1

Fig. 6. Tag setup in KepServerEX using OPC UA.

- Both PLCs in WinCC Explorer provide one screen that makes it easy to watch the entire system. Workers can change the conveyor speed, observe how the motor condition trends go, and react to alarms immediately. Therefore, the suggested SCADA link can significantly improve system visibility, enable proactive maintenance strategies, and improve efficiency.

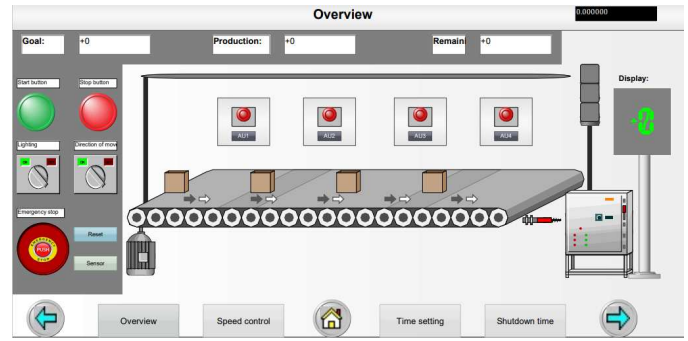


Fig. 7. OPC UA tag setup in KepServerEX..

C. MQTT-Based Remote Monitoring

To provide a means for remote data processing, an MQTT-based communication design was executed using a Raspberry Pi board and a Node-RED tool. Thus, this setup provides real-time data exchange between any PLC and the mobile monitoring interface under low-latency conditions. The Raspberry Pi was configured as an MQTT broker and used to receive real-time information from Siemens and Mitsubishi PLCs. These PLCs published important information regarding the process, such as the conveyor status, motor temperature, and vibration levels. This information can be subscribed to by remote clients such as SCADA, mobile devices, or any web-based dashboard. MQTT provides lightweight, secure, and efficient data transmission, which is suitable for industrial applications.

To set up a Raspberry Pi as an MQTT broker, we followed the following steps:

- **Installation of Mosquitto:** The first step in the entire procedure is the installation of Mosquitto, which refers to MQTT broker software. This is performed by executing command line instructions inside the terminal.
- **Setting up Node-RED:** Mosquito is installed on the Raspberry Pi, followed by the installation of the Node-RED. With node-REDs, the orchestration and management of data flows become fairly easy.
- **Data Acquisition:** Configuration of Raspberry Pi is performed in such a way that it can acquire data from the Programmable Logic Controller (PLC) Siemens S7-1200 model and node integration for related information extraction from the PLC via the Node-RED library.
- **Data Publishing:** The acquired data were published using the MQTT protocol in Mosquitto. This is an important step in real-time communication, which allows data to be sent to distant subscribers.
- **MQTT Configuration:** Major parameters are configured, including QoS settings for reliable message delivery. This QoS defines how messages are sent and received to

optimize performance and reliability. An external client (such as the IO Adafruit website) was used to test this setup and monitor the published data to ensure that everything works well.

- **Remote Access:** Finally, remote access instructions for the Raspberry Pi are provided to the SSH used for secure connection, allowing monitoring and management from other devices.

A personalized node-RED dashboard was created on Raspberry Pi, providing an easy web-based face for watching from afar. Subsequently, the dashboard can show real-time motor heat and shake levels; it will also show the assembly line status, conveyor speed changes, and emergency stop warnings. People can view the dashboard using a phone or web browser, making sure to process the data watch from any place [12, 13]. A screenshot of the node-RED dashboard displaying live PLC data is shown in Fig. 8.

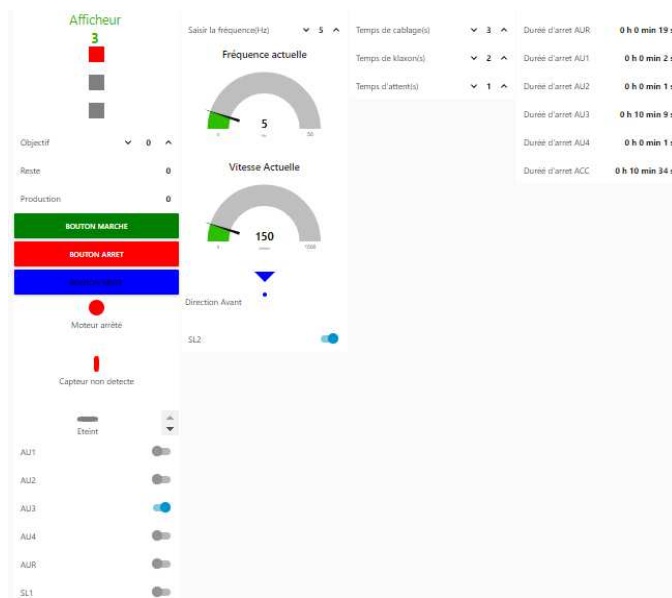


Fig. 8. Node-RED dashboard displaying live PLC data.

IV. RESULTS AND DISCUSSION

The multi-PLC automated system significantly improved process control, monitoring, and remote access to the environment. With the integration of WinCC Explorer, data from both Siemens and Mitsubishi PLCs were unified to present real-time sensor readings and process variables using a single interface. This centralized visibility makes operations more efficient and improves decision-making.

The setup also afforded better monitoring and predictive maintenance for virtually all assembly operations and motor-health tracking. Fault detection through the Mitsubishi PLC facilitates the early identification of issues that could lead to critical failures; hence, unexpected downtimes can be avoided.

The OPC UA (for SCADA integration) was used here to ensure seamless data exchange along with MQTT (for remote

monitoring), which provides strong and effective communication over the entire system.

For remote monitoring, this approach improved flexibility and added IoT integration but also brought in cybersecurity risks like: Unauthorized access (data may get manipulated or operations disrupted), Data interception (due to unencrypted TCP/IP communication), Denial-of-Service DoS attack system availability

Malware infiltration risks (if the RPI is linked to larger networks). Curing these weaknesses is the key to making the system's long-run trustworthy and safe. Future improvements should include tough authentication parts, encryption rules, and network splitting to reduce these dangers [14].

V. CONCLUSION

This study proposes design procedures to demonstrate a multi-PLC auto-mation setup that combines SCADA, HMI, and IoT technologies. This offers real-time control over processes, along with predictive maintenance and remote monitoring. Specifically, the WinCC Explorer SCADA system was demonstrated to efficiently oversee both Siemens S7-1200 (control of assembly line) and Mitsubishi FX3GE (monitoring motor condition) from a single unified centralized interface; data interoperability was also ensured through OPC UA seam-less information exchange. Furthermore, MQTT plus Node-RED joined mobile access for key process parameters, thus enhancing the system flexibility together with remote diagnostics. The designed system results indicate that a reduction in unexpected downtimes, improved maintenance strategies, and increased system efficiency could further include the integration of a lightweight intrusion detection system (IDS) implementing mutual certificate-based authentication as well as containerizing the broker to further isolate and protect the communication stack. These enhancements can boost the security compliance of systems in industrial environments that require secure IoT architectures.

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